Applied Physics Letters

A NOTE ON THE MAXIMUM CRITICAL FIELD OF HIGHFIELD SUPERCONDUCTORS

B. S. Chandrasekhar

Citation: Appl. Phys. Lett. **1**, 7 (1962); doi: 10.1063/1.1777362
View online: http://dx.doi.org/10.1063/1.1777362
View Table of Contents: http://apl.aip.org/resource/1/APPLAB/v1/i1
Published by the American Institute of Physics.

ACKNOWLEDGMENTS

The assistance of J. J. Schott, who prepared the specimen for this investigation, is gratefully acknowledged.

[1] A. L. Jain, *Phys. Rev.*, **114**, 1518 (1959).

[2] G. E. Smith and R. Wolfe, *J. Appl. Phys.* **33**, 841 (1962).

[3] R. Wolfe and G. E. Smith, *Bull. Amer. Phys. Soc. II* **7**, 173 (1962).

[4] J. R. Drabble and R. Wolfe, *J. Electronics and Control* 3, 259 (1957).

[5] B. Abeles and S. Meiboom, *Phys. Rev.* **115**, 1561, (1959).

[6] C. Herring, T. H. Geballe, and J. E. Kunzler, *Phys. Rev.* **111**, 36 (1958).

[7] A. E. Bowley, R. Delves, and H. J. Goldsmid, *Proc. Phys. Soc.* **72**, 401 (1958).

A NOTE ON THE MAXIMUM CRITICAL FIELD OF HIGH-FIELD SUPERCONDUCTORS

B. S. Chandrasekhar

Westinghouse Research Laboratories, Pittsburgh, Pennsylvania

(Received July 23, 1962)

A fair number of superconductors, mostly alloys of the transition metals, are now known to exhibit zero resistance in applied magnetic fields up to 100 kG or more.[1] It has become increasingly evident, mainly as a result of the work of Berlincourt and Hake,[1] that the maximum field H_m at which a given alloy still exhibits superconductivity (as indicated by zero resistance for current densities less than 10 A/cm^2) is a true parameter of the alloy, independent of its degree of cold work. They have also shown that H_m peaks sharply between 4 and 5 electrons/atom, reaching for example a value of ~145 kG for Ti-Nb.

The existence of superconductivity at very high fields is probably due, as has been suggested by several authors,[2,3] to a "filamentary" structure, with at least one characteristic dimension small compared to the penetration depth. But this model does not explain the dependence of H_m on the electron/atom ratio, which is strongly reminiscent of the dependence of T_c on the same ratio.[4] The model also does not suggest a natural upper limit for H_m for a given material.

A mechanism is proposed here which does suggest such a limit for H_m and also leads to a relation between H_m and T_c. It is based on the fact that in these filamentary superconductors there is practically complete penetration of the superconducting regions by the magnetic field. One would therefore expect that when the magnetic energy $2\mu H$ (where μ = the Bohr magneton) becomes comparable to the energy gap Δ ($\sim 3.5\, kT_c$ at low temperatures) superconductivity will be destroyed. This leads to the relation $H_m \cong 2.6 \times 10^4\, T_c$. It is immediately seen that this relation (1) gives agreement in order of magnitude with experimentally determined values of H_m (see Table 1; Mo_3Re is an exception and is referred to later); (2) predicts the relation between H_m and T_c in a given alloy system, mentioned above; and (3) suggests that H_m is roughly proportional to T_c even for different alloy systems and compounds. This last suggestion agrees with the observation that the alloys with T_c in the range 10-12°K have values of H_m which are roughly half the value estimated for Nb_3Sn with a T_c of about 18°K.

Table 1. Experimental values of T_c and H_m

	Nb_3Sn	Nb-Zr	Nb-Ti	Mo_3Re
T_c(°K)	18	10-12	9.5	11
H_m(kG)	200-300 (estimated)	70-90	145	15

It must be emphasized that the above argument only sets an upper limit for H_m; the actual value observed for a given material may be limited by specific metallurgical factors which determine the size of the superconducting filaments. Thus Mo_3Re with a T_c of ~11°K shows a rather low H_m of 15 kG.[2] Further, the orbital magnetic energy of the electrons (as also the field dependence of the energy gap)[5] has been neglected. The purpose of this note is only to point out that this effect of the magnetic field on the electron spin, which has hitherto been considered in the various theoretical attempts[6,8] to explain the Knight shift in superconductors, must also be included in any detailed theory of high-field superconductors. On the basis of the above model, one would expect $V_3Ga(T_c = 16.8°K)$ to have an H_m not much different from that of Nb_3Sn; the estimates ranging to 700 kG extrapolated from measurements close to T_c may turn out to be optimistic.

Interesting discussions with T. G. Berlincourt and P. G. Harper are acknowledged.

[1] T. G. Berlincourt and R. R. Hake, *Bull. Am. Phys. Soc. II* **7**, 408 (1962).
[2] J. E. Kunzler, *Rev. Mod. Phys.* **33**, 501 (1961).
[3] J. J. Hauser and E. Helfand, to be published.
[4] J. K. Hulm and R. D. Blaugher, *Phys. Rev.* **123**, 1469 (1961).
[5] D. H. Douglass, Jr., *IBM Journal of Res. and Dev.* **6**, 44 (1962).
[6] V. Heine and A. B. Pippard, *Phil. Mag.* **3**, 1046(1958).
[7] K. Yosida, *Phys. Rev.* **110**, 769 (1958).
[8] L. N. Cooper, *Phys. Rev. Letters* **8**, 367 (1962).

LOW-TEMPERATURE INTERNAL FRICTION PEAKS IN GOLD

G. A. Aler, R. E. Condit, and J. A. Karbon
Scientific Laboratory, Ford Motor Company, Dearborn, Michigan

(Received July 12, 1962)

It is well known that in fcc metals cold work in the form of a few percent plastic deformation gives rise to peaks in the curve of internal friction vs temperature at low temperatures.[1] Theoretical interpretations of this phenomenon in terms of dislocation motion have been given by several authors.[2,3] Both the theory and the experiments characterize the peaks in terms of an activation energy which is determined by observing the temperature at which the peak occurs as a function of measurement frequency. In connection with some other experiments the locations of two internal friction peaks in gold have been determined at 18 kilocycles. The position of the high-temperature peak at 121°K is in reasonable agreement with the recent results of Bordoni, Nuovo, and Verdini.[4] The low-temperature peak located at 70°K is outside the temperature range covered by Bordoni *et al.*,[4] so the present results represent the first measurements on this peak at kilocycle frequencies. From Okuda's[5] measurements on both peaks at ½ cps, an activation energy for the process giving rise to the peaks may be estimated at 0.2 ev and 0.1 ev for the high- and low-temperature peaks respectively. These activation energies should be considered as only approximate since it has been pointed out[1-3] that the same cold-working procedure should be used on the specimens for all frequencies. It is interesting to note that these activation energies are approximately twice the values obtained on copper even though the two metals have nearly the same melting points and the Debye temperature of gold is roughly half that of copper.

The measurements of Q^{-1} shown in Fig. 1 were obtained by measuring the width of the curve of vibrational amplitude vs frequency around the fundamental longitudinal resonant frequency of a ¼-in.-diam x 2¼-in.-long bar. An eddy-current drive at one end of the bar induced the longitudinal vibrations, while a modified capacity micrometer[6] was used at the other end to measure the amplitude of vibration. The modification to the capacity micrometer consisted in placing a pancake coil of wire